

From Scalar Residual Injection to Iteration-Wise PA-Model Adjoint Updates

A Mathematical Note on Waveform Digital Predistortion

August 2026

Abstract

This note formulates waveform iterative learning control (ILC) for digital predistortion (DPD) as a constrained inverse problem. It shows that scalar residual injection implicitly replaces the local power-amplifier (PA) sensitivity by a scaled identity, derives the resulting convergence and descent limitations, and then formulates an iteration-wise forward-model alternative. The measured residual is backpropagated through a frozen real-linear PA-model adjoint; a damped Gauss-Newton or Levenberg-Marquardt step supplies regularized local inversion. The presentation is theoretical and contains no experimental claims.

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1 Scope and notation

Throughout this note, *classical scalar ILC* denotes the waveform update that directly injects a scaled output residual into the PA input. This terminology does not cover the full lifted-domain ILC literature, in which a designed learning operator, robustness filter, identified plant inverse, or instantaneous-gain preconditioner may be used. The limitation studied here is therefore not repetition-domain learning itself, but the scalar identity-sensitivity approximation [1, 2].

Let

- $u \in \mathbb{C}^N$ be the finite waveform applied to the PA;
- $F : \mathbb{C}^N \rightarrow \mathbb{C}^N$ be the calibrated PA input-output operator;
- $d \in \mathbb{C}^N$ be the desired PA output;
- $y_k = F(u_k)$ be the measured output at outer iteration k ; and
- $r_k = d - y_k$ be the output tracking residual.

Some implementation descriptions instead use $e_k = y_k - d = -r_k$. This note retains $r_k = d - y_k$ throughout, so the classical update has the familiar plus-residual form.

The waveform-level inverse problem is

$$\min_{u \in \mathcal{U}} \mathcal{L}(u), \quad \mathcal{L}(u) = \frac{1}{2} \|F(u) - d\|_2^2 = \frac{1}{2} \|r(u)\|_2^2, \quad (1)$$

where $\mathcal{U}$ is the feasible set imposed by peak, RMS, PAPR, and hardware limits.

The derivation assumes a fixed input/output coordinate system. If the measured output is independently renormalized at every iteration, the effective operator is an iteration-dependent composition of the PA and the calibration map rather than the physical PA map alone.

A nonlinear RF PA is generally not holomorphic as a complex map because its basis functions depend on $|u|$. We therefore treat $\mathbb{C}^N$ as $\mathbb{R}^{2N}$, with real inner product

$$\langle a, b \rangle_{\mathbb{R}} = \text{Re}(a^H b). \quad (2)$$

Let $J(u)$ denote the real-linear Fréchet derivative of F :

$$F(u + h) = F(u) + J(u)h + o(\|h\|_2). \quad (3)$$

Its real adjoint $J(u)^T$ is defined by

$$\langle v, J(u)h \rangle_{\mathbb{R}} = \langle J(u)^T v, h \rangle_{\mathbb{R}}. \quad (4)$$

Because $r(u) = d - F(u)$, the directional derivative of the loss is

$$D\mathcal{L}(u)[h] = -\langle r(u), J(u)h \rangle_{\mathbb{R}} = \langle -J(u)^T r(u), h \rangle_{\mathbb{R}}. \quad (5)$$

Hence the true input-space gradient is

$$\boxed{\nabla \mathcal{L}(u) = -J(u)^T r(u)}. \quad (6)$$

The output residual is therefore not, in general, an input correction. It must be transported through the PA adjoint before it becomes a gradient in the input coordinates.

2 Classical scalar ILC as an identity-sensitivity approximation

The scalar linear ILC update is

$$u_{k+1} = u_k + \mu r_k, \quad \mu > 0. \quad (7)$$

Using a first-order PA expansion at u_k gives

$$\begin{aligned} r_{k+1} &= d - F(u_k + \mu r_k) \\ &= r_k - \mu J_k r_k + o(\mu \|r_k\|_2), \end{aligned} \quad (8)$$

where $J_k = J(u_k)$. Thus

$$r_{k+1} \approx (I - \mu J_k) r_k. \quad (9)$$

Equation (7) implicitly treats the output residual as if it were already expressed in the PA input coordinates. Equivalently, it approximates the local inverse learning operator, or the adjoint sensitivity after step scaling, by a scalar multiple of the identity.

2.1 Linear and scalar convergence conditions

For a fixed linear PA $F(u) = Au$, asymptotic convergence requires

$$\rho(I - \mu A) < 1, \quad (10)$$

where $\rho(\cdot)$ is the spectral radius. For a nonzero scalar complex gain $A = g$, a real positive learning rate must satisfy

$$|1 - \mu g| < 1 \iff 0 < \mu < \frac{2\operatorname{Re}(g)}{|g|^2}. \quad (11)$$

Such a μ exists only when $\operatorname{Re}(g) > 0$. A sufficiently large PA phase rotation can therefore turn the nominal correction into a non-contractive direction. For example, if $g = j$, then $|1 - j\mu| > 1$ for every $\mu > 0$.

The same conclusion follows directly from the loss derivative. Along the classical step $h = \mu r_k$,

$$D\mathcal{L}(u_k)[\mu r_k] = -\mu \langle r_k, J_k r_k \rangle_{\mathbb{R}}. \quad (12)$$

The step is a first-order descent direction only if

$$\langle r_k, J_k r_k \rangle_{\mathbb{R}} > 0. \quad (13)$$

This condition is not guaranteed for a general complex, nonlinear, or dynamic PA.

2.2 Main limitations

1. **Direction error.** Phase distortion or negative local AM/AM slope can make $\langle r_k, J_k r_k \rangle_{\mathbb{R}} \leq 0$. Identity transport can then be orthogonal to, or point against, the true descent direction.
2. **Scale error.** If the local PA slope is small, the output changes little although μr_k continues to accumulate at the input. If the slope is large, the same fixed μ can overshoot. One scalar cannot simultaneously invert widely different singular directions.
3. **Memory and cross-sample coupling.** With PA memory, J_k is not diagonal. An error at one time index may require corrections at several input indices. Identity transport ignores this coupling.
4. **Iteration-dependent nonlinearity.** The Jacobian changes with u_k . A learning rate that is stable in one envelope region need not remain stable after the waveform moves to another region.
5. **Saturation and infeasibility.** In deep saturation, J_k may become rank deficient or develop very small singular values. Scalar ILC still injects μr_k along locally insensitive directions. If $d \notin F(\mathcal{U})$, then, for compact $\mathcal{U}$ and continuous F ,

$$\min_{u \in \mathcal{U}} \|F(u) - d\|_2 > 0. \quad (14)$$

No ILC law can drive such an error to zero.

6. **Noise integration.** If $y_k = F(u_k) + n_k$, the classical update directly contains $-\mu n_k$. Repetition can accumulate capture noise unless averaging, filtering, regularization, or a stopping rule is used.

These are limitations of the identity-Jacobian baseline, not impossibility statements about every form of ILC. A known plant inverse, instantaneous complex gain, or frequency-dependent learning filter can correct some of them when its structural assumptions hold.

2.3 Instantaneous gain as a diagonal secant preconditioner

For nonzero input samples, instantaneous-gain ILC estimates a pointwise secant gain

$$\hat{g}_k[n] = \frac{y_k[n]}{u_k[n]}, \quad \widehat{D}_k = \operatorname{diag}(\hat{g}_k[n]). \quad (15)$$

A damped diagonal update is

$$\delta_k = (\widehat{D}_k^H \widehat{D}_k + \lambda I)^{-1} \widehat{D}_k^H r_k, \quad \lambda > 0. \quad (16)$$

This is a diagonal secant preconditioner, not the full differential inverse of a general nonlinear PA. To see the distinction, let $f(u) = a(\varrho)u$ with $\varrho = |u| > 0$. Its real-linear differential is

$$Df(u)[h] = \left(a(\varrho) + \frac{\varrho}{2} a'(\varrho) \right) h + \frac{a'(\varrho)u^2}{2\varrho} \bar{h}. \quad (17)$$

The ratio $y/u = a(q)$ omits the amplitude-dependent differential terms, the conjugate path, and all cross-sample memory coupling. Low-amplitude division also requires thresholding. Nevertheless, instantaneous gain can be highly effective for approximately memoryless multiplicative distortion and is not assumed to be weaker than the forward-model method [1, 6].

3 Iteration-wise forward PA modeling

Backpropagation through a fitted forward PA model has established precedents [4, 5]. In the waveform-level construction considered here, the model-adjoint scheme estimates the missing local PA geometry from the latest measured pair (u_k, y_k) and refits a differentiable forward model at every outer ILC iteration:

$$\hat{F}_k(u) = \hat{F}(u; \hat{\theta}_k), \quad (18)$$

for example by regularized regression,

$$\hat{\theta}_k = \arg \min_{\theta} \|\hat{F}(u_k; \theta) - y_k\|_2^2 + \gamma \|\theta\|_2^2. \quad (19)$$

The model regularization γ is distinct from the update damping introduced later.

The fitted parameters $\hat{\theta}_k$ and envelope scale are frozen while iteration k 's update is computed. Gradients are not propagated through the regression solver, measurement process, envelope scale, delay estimator, or calibration estimator. The model is refitted only after a new PA input-output pair has been observed.

This yields two nested loops:

1. an *identification loop*, which estimates a local forward PA model from the current measurement;
2. a *learning loop*, which uses the frozen model derivative to transport the measured residual back to the PA input.

The residual remains the physical measurement residual $r_k = d - y_k$; it is not replaced by a model residual.

4 Real-linear backpropagation through the PA model

Let

$$\hat{J}_k = D\hat{F}_k(u_k) \quad (20)$$

be the frozen model derivative. Its Jacobian-vector product (JVP) maps an input perturbation to an output perturbation, $h \mapsto \hat{J}_k h$, and its vector-Jacobian product (VJP) transports an output cotangent back to the input, $v \mapsto \hat{J}_k^T v$.

4.1 Memory-polynomial example

Consider a circular memory polynomial [3]

$$\hat{F}_k(u)[n] = \sum_{m=0}^{M-1} \sum_{p \in \mathcal{P}} c_{p,m} z_m[n] \left(\frac{|z_m[n]|}{s} \right)^{p-1}, \quad z_m[n] = u[(n-m) \bmod N]. \quad (21)$$

Here $\mathcal{P} \subset \{1, 3, 5, \dots\}$ is a set of positive odd orders. For one basis function

$$\phi_p(z) = c z \left(\frac{|z|}{s} \right)^{p-1}, \quad (22)$$

the real-linear differential contains both direct and conjugate terms:

$$D\phi_p(z)[h] = a_p(z)h + b_p(z)\bar{h}, \quad (23)$$

where

$$a_p(z) = c \frac{p+1}{2} \left(\frac{|z|}{s} \right)^{p-1}, \quad (24)$$

$$b_p(z) = c \frac{p-1}{2} \left(\frac{z}{s} \right)^2 \left(\frac{|z|}{s} \right)^{p-3}, \quad p > 1. \quad (25)$$

For $p = 1$, define $b_1(z) = 0$ explicitly; at $z = 0$, use the continuous extension $b_p(0) = 0$. After summing polynomial orders at each delay into arrays a_m and b_m , the JVP is

$$\hat{J}_k h = \sum_m \left[a_m \odot \text{roll}(h, m) + b_m \odot \overline{\text{roll}(h, m)} \right]. \quad (26)$$

The corresponding real adjoint is

$$\hat{J}_k^T v = \sum_m \text{roll}(\overline{a_m} \odot v + b_m \odot \bar{v}, -m). \quad (27)$$

The conjugate contribution in Eq. (27) is $b_m \odot \bar{v}$, not $\overline{b_m} \odot \bar{v}$. An ordinary complex-linear Hermitian derivative would omit this path and generally produce the wrong gradient.

5 Backpropagated learning laws

5.1 Raw model-adjoint update

Replacing identity transport by the learned adjoint gives

$$\delta_k^{\text{VJP}} = \eta \hat{J}_k^T r_k, \quad u_{k+1} = u_k + \delta_k^{\text{VJP}}, \quad \eta > 0. \quad (28)$$

If $\hat{J}_k = I$, this reduces exactly to scalar linear ILC with $\eta = \mu$. With an exact derivative, $\hat{J}_k = J_k$, the first-order loss change is

$$D\mathcal{L}(u_k)[\delta_k^{\text{VJP}}] = -\eta \|J_k^T r_k\|_2^2 < 0 \quad (29)$$

whenever the true gradient is nonzero. This negative directional derivative guarantees actual decrease only for a sufficiently small finite step; it is not a global or arbitrary-step guarantee.

With derivative error $J_k = \hat{J}_k + E_k$,

$$D\mathcal{L}(u_k)[\delta_k^{\text{VJP}}] = -\eta \langle J_k^T r_k, \hat{J}_k^T r_k \rangle_{\mathbb{R}}. \quad (30)$$

Here $\|E_k\|_2$ is the operator 2-norm induced by the $2N \times 2N$ real representation of the real-linear map E_k . Furthermore,

$$\langle J_k^T r_k, \hat{J}_k^T r_k \rangle_{\mathbb{R}} \geq \|\hat{J}_k^T r_k\|_2 (\|\hat{J}_k^T r_k\|_2 - \|E_k\|_2 \|r_k\|_2). \quad (31)$$

A sufficient local descent condition is therefore

$$\|E_k\|_2 \|r_k\|_2 < \|\hat{J}_k^T r_k\|_2. \quad (32)$$

The learned adjoint is thus a surrogate gradient. Backpropagation through a fitted model does not by itself establish that the direction equals the true PA gradient.

5.2 Damped Gauss-Newton / Levenberg-Marquardt update

Raw gradient descent still inherits poor Jacobian scaling. A regularized local step instead solves

$$\delta_k = \arg \min_{\delta} \frac{1}{2} \|r_k - \hat{J}_k \delta\|_2^2 + \frac{\lambda_k}{2} \|\delta\|_2^2, \quad \lambda_k > 0. \quad (33)$$

Its first-order condition is the damped normal equation

$$\boxed{(\hat{J}_k^T \hat{J}_k + \lambda_k I) \delta_k = \hat{J}_k^T r_k.} \quad (34)$$

The normal operator is positive definite in the real geometry because, for every nonzero q ,

$$\langle q, (\hat{J}_k^T \hat{J}_k + \lambda_k I) q \rangle_{\mathbb{R}} = \|\hat{J}_k q\|_2^2 + \lambda_k \|q\|_2^2 > 0. \quad (35)$$

Consequently, the step can be computed by matrix-free conjugate gradients using only

$$q \mapsto \hat{J}_k q \mapsto \hat{J}_k^T (\hat{J}_k q) + \lambda_k q, \quad (36)$$

without constructing the full $2N \times 2N$ real Jacobian. The same frozen linearization must be used within one CG solve so that the normal operator remains symmetric positive definite.

If $\hat{J}_k = U \Sigma V^T$ is the singular-value decomposition of its real representation, then

$$\delta_k = V \operatorname{diag}\left(\frac{\sigma_i}{\sigma_i^2 + \lambda_k}\right) U^T r_k. \quad (37)$$

Well-observed directions are approximately inverted, whereas weak directions are suppressed. Indeed,

$$0 \leq \frac{\sigma}{\sigma^2 + \lambda_k} \leq \frac{1}{2\sqrt{\lambda_k}}. \quad (38)$$

Two limiting cases clarify the connection to simpler ILC laws:

- If $\hat{J}_k = I$, then $\delta_k = r_k / (1 + \lambda_k)$, a damped scalar ILC step.
- For a memoryless complex gain g ,

$$\delta_k = \frac{g^*}{|g|^2 + \lambda_k} r_k, \quad (39)$$

which is a Tikhonov-regularized complex-gain inverse.

For a perfect linear model $F(u) = Au$, the new residual is

$$r_{k+1} = [I - A(A^T A + \lambda I)^{-1} A^T] r_k. \quad (40)$$

Along each nonzero singular direction, the contraction factor is

$$\frac{\lambda}{\sigma_i^2 + \lambda}. \quad (41)$$

With $\lambda = 0$ and invertible A , the exact linear inverse is recovered in one idealized step.

6 Measurement-anchored prediction and safeguards

A forward model can have static bias at the current waveform. Using $\hat{F}_k(u_k + \delta)$ directly would mistake that bias for a measured output change. Instead use the anchored prediction

$$\tilde{y}_k(\delta) = y_k + \hat{F}_k(u_k + \delta) - \hat{F}_k(u_k). \quad (42)$$

It satisfies $\tilde{y}_k(0) = y_k$ exactly while retaining the model-predicted nonlinear increment. A candidate can be accepted only when, for example,

$$\frac{1}{2} \|d - \tilde{y}_k(\delta)\|_2^2 < \frac{1}{2} \|r_k\|_2^2. \quad (43)$$

This is a predicted-decrease condition under the model, not a guarantee of monotone decrease for the physical PA.

The update should additionally obey a trust region and the feasible input set:

$$\operatorname{RMS}(\delta) \leq \tau \operatorname{RMS}(u_k), \quad u_k + \delta \in \mathcal{U}. \quad (44)$$

If the full step is rejected, test $\beta^b \delta$, $0 < \beta < 1$, until a candidate passes the anchored decrease and input constraints. If none passes, hold u_k . A model-fit failure should likewise produce an explicit hold or declared scalar-ILC fallback rather than silently reusing an unrelated model.

These safeguards bound the consequences of local model error. They do not create an inverse in saturation, guarantee global convergence, or ensure branch switching in a nonmonotone PA.

7 Saturation, reachability, and remaining limitations

The learned model changes the update from identity transport,

$$\delta_k = \mu r_k, \quad (45)$$

to model-adjoint transport,

$$\delta_k = \eta \hat{J}_k^T r_k, \quad (46)$$

or a regularized local inverse, Eq. (34). This directly addresses phase rotation, unequal local slope, and memory coupling when the forward model is locally accurate. Iterative refitting lets the fitted derivative track the waveform through the PA operating region.

The method remains local and model-dependent:

- it has no global convergence guarantee for a nonconvex PA inverse;
- model mismatch can rotate the learned gradient away from the true gradient;
- a poorly excited waveform cannot identify unobserved model directions;
- changing calibration across iterations changes the effective operator being learned;
- capture noise affects both the residual and the fitted model;
- the optimized variable is one finite, repeatable waveform, so arbitrary traffic requires a separate parameterized DPD identification stage; and
- if $d \notin F(\mathcal{U})$, no gradient or second-order approximation can make the target reachable.

In saturation, J_k may become rank deficient. A residual component in $\ker(J_k^T)$ cannot be reduced to first order. In the extreme locally constant idealization $J_k = 0$, if the learned model faithfully represents that behavior, then $\hat{J}_k = 0$ and the right-hand side of Eq. (34) vanishes. Positive damping gives $\delta_k = 0$. This is a safe indication of local unresponsiveness, not recovery of the unreachable desired waveform. Conversely, a small learned gradient can also result from model mismatch, so a local saturation indicator is not a proof of global unreachability.

8 Compact algorithm and interpretation

At each outer iteration k :

1. apply u_k , measure and align y_k , and form $r_k = d - y_k$;
2. fit and validate $\hat{F}_k$ from the current input-output pair;
3. freeze $\hat{F}_k$ and construct its real-linear JVP and VJP at u_k ;
4. compute a raw VJP step or solve Eq. (34) by matrix-free CG;
5. apply trust-region scaling, input projection, anchored nonlinear prediction, and backtracking;
6. accept the first safe predicted-decrease candidate, otherwise hold or use a declared fallback; and
7. refit only after the next physical PA measurement.

The conceptual improvement is not merely the addition of a model. It replaces the identity-sensitivity approximation with the differential of an iteration-wise fitted forward model, uses the real adjoint to transport the measured residual to the PA input, and regularizes that transport before the next ILC experiment.

This construction combines established PA modeling, backpropagation, and ILC ingredients in a specific waveform-level loop. It does not claim the first use of PA forward models, PA Jacobians, memory backpropagation, direct learning, memory-polynomial modeling, or model-based ILC. Here “improved” means that the update contains richer local sensitivity information and explicit numerical safeguards; it does not mean uniform superiority over instantaneous-gain, BLA-inverse, or other plant-informed ILC methods.

Companion documents

The following repository documents provide the implementation and prior-art context:

- algorithm conventions: docs/algorithm_design.md;
- prior-art boundaries: docs/prior_art_review.md; and
- Markdown edition: docs/ilc_dpd_forward_model_theory.md.

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